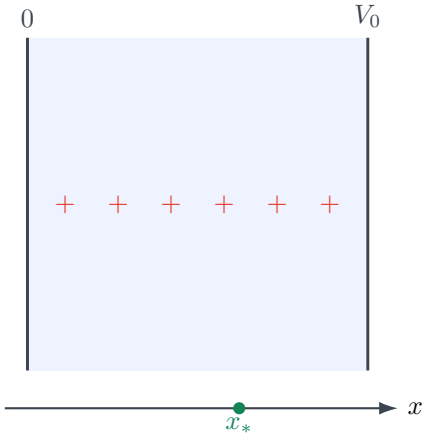
均匀带电平板层：线性边界势与体电荷抛物项共同决定零场点



$$\varphi = \frac{V_0 x}{L} + \frac{\rho_0}{2\varepsilon_0} x(L - x)$$

$$E_x = \frac{\rho_0}{\varepsilon_0}\left(x - \frac{L}{2}\right) - \frac{V_0}{L}$$

$x_* = \dfrac{L}{2} + \dfrac{\varepsilon_0 V_0}{\rho_0 L}$；仅当 $0 < x_* < L$ 时零场点位于层内。